

Analytical Toolkit and Modeling Innovations

Susanna Grigoryan

Recent Staff Member, Monetary Policy Department, Central Bank of Armenia

Student, Global Forecasting School

November 2025

Note

1. Limitations of FPAS Mark I

FPAS Mark I relied on a deterministic baseline and a single linear core model. While useful for transparency, it could not capture nonlinearities, asymmetric risks, credibility dynamics, or rapid shifts in financial conditions. These limitations motivated the development of FPAS Mark II, which is designed to better support policy under uncertainty.

Since the onset of the COVID-19 pandemic, the global policy environment has been characterized by a level of uncertainty that has become—using Alan Greenspan’s words—the “defining feature of the monetary policy landscape.” A series of overlapping shocks, including geopolitical tensions, energy-price volatility, and supply-chain disruptions, has reinforced this uncertainty and exposed the limits of relying on a single baseline projection. The world economy has become more vulnerable to nonlinear dynamics and high-impact tail events, prompting a fundamental reassessment of how central banks forecast and analyze the economy.

These developments were a major catalyst for the design of FPAS Mark II and also highlighted the need to upgrade the analytical tools that support monetary-policy analysis. A modern framework must be equipped to construct scenarios, test sensitivities, and examine the credibility implications of alternative policy paths.

The earlier generation of FPAS, developed by Berg, Karam, and Laxton (2006), transformed monetary-policy analysis across central banks. Its core tool, the Quarterly Projection Model (QPM), provided a structured, accessible way to support inflation-targeting regimes. QPMs rely on New Keynesian principles—nominal and real rigidities, business-cycle behavior, and rational expectations—combined with a policy rule that guides interest-rate adjustments to stabilize inflation and anchor expectations. The CBA

adopted this approach in 2006, in line with many central banks transitioning to inflation targeting.

However, these conventional frameworks have important limitations. They rely on linear relationships that may break down during periods of economic stress, and they treat credibility as fixed and external to the model. In reality, credibility evolves with policy performance and can deteriorate quickly. Ignoring these nonlinearities creates risks. For example, when the output gap becomes strongly positive, the Phillips Curve may steepen, causing inflation expectations to accelerate more rapidly than a linear model would suggest. Failing to recognize this can lead to delayed policy action, requiring more aggressive tightening later and resulting in larger economic costs. The same logic applies to credibility: it can erode suddenly but is difficult and costly to rebuild.

In a world of heightened uncertainty and competing interpretations of the data, an analytical framework that captures these nonlinearities becomes essential. It allows policymakers to better understand the full consequences of potential policy missteps and to make decisions that minimize regrets. For this reason, nonlinear analytical structures—and the risk-management perspective they support—are a core component of the Prudent Risk Management Approach to Price Stability.

2. Nonlinearities and State Dependence

FPAS Mark II incorporates nonlinear relationships—particularly in the Phillips Curve, the transmission mechanism, and credibility dynamics. Inflation responds more strongly in overheating conditions than in slack periods, implying that late tightening is more costly than early measured action. Nonlinearities help policymakers model real-world behaviour more accurately and capture asymmetric risks inherent in modern monetary policy.

3. Endogenous Credibility

FPAS Mark II models credibility as an evolving stock influenced by central banks ability to meet its targets, its communication quality, and institutional consistency. Expectations adjust as the central bank demonstrates competence and reliability. Modelling credibility endogenously strengthens the connection between current policy actions and future outcomes, recognizing that credibility must be earned continuously.

4. The FPAS Mark II Model Suite

FPAS Mark II employs a suite of complementary models rather than relying on a single structure. These include:

- EndoCred Model: Incorporates endogenous credibility, nonlinear Phillips Curve dynamics, and asymmetric loss functions.
- MPMOD: Informs estimates of latent variables, such as the output gap, with theoretical relationships linking unobservable with observable variables.
- FCMOD: Captures financial cycle dynamics and systemic vulnerabilities.
- Linear-QPM: Retained for continuity, historical decompositions, and pedagogical comparison.
- Monthly Projection Model: Used for thinking of economic scenarios using data that is available on more frequent basis.
- Dynamic Stochastic General Equilibrium Models (DSGE): Examples are GIMF-type models used for medium-term policy analysis, especially when fiscal and monetary interactions matter.
- GPM: 3-country (US, EU, Russia) plus commodities (oil, copper, food) linear model used for analyzing the global sector

Using a model suite helps diversify analytical risks and provides multidimensional insight into inflation, output, and financial conditions.

ENDOCRED Model

The ENDOCRED model builds on the foundations of the traditional QPM introduced by Berg, Karam, and Laxton (2006), while introducing several critical innovations suited to a world of heightened and persistent uncertainty. A central feature of the model is the endogenization of policy credibility. Instead of treating credibility as fixed, ENDOCRED allows it to evolve over time in a nonlinear way, directly influenced by the central bank's past performance. This dynamic structure captures the reality that credibility can erode quickly when policy missteps occur and strengthens only gradually when policy is consistent and disciplined.

Inflation dynamics are modeled through a convex Phillips curve, which responds more strongly when inflation is high or when the output gap becomes significantly positive. This specification reflects real-world behavior more accurately than a linear formulation, particularly during periods of overheating or when expectations begin to de-anchor.

The IS curve is also extended beyond its traditional short-term focus. ENDOCRED introduces a long-term effective interest rate derived from yields across different maturities based on the expectations hypothesis and a time-varying term premium. This feature allows the model to incorporate the broader and more persistent influence of monetary policy on economic activity, capturing the role of the yield curve in shaping demand conditions.

In contrast to the standard Taylor-type policy rule used in many QPMs, ENDOCRED adopts a loss-function-based policy design. This enables more flexible, state-contingent responses, which are essential when uncertainty is elevated, transmission is impaired, or the economy is operating near policy limits.

Another key innovation is the explicit treatment of dollarization. The influence of external interest rates on the domestic effective interest rate depends on a dollarization parameter that is inversely related to policy credibility. As credibility improves, dollarization declines, strengthening the effectiveness of domestic monetary policy. This creates a meaningful feedback loop: stronger credibility reduces dollarization, which enhances transmission, which in turn helps credibility rise further.

Together, these innovations allow ENDOCRED to capture nonlinearities, credibility dynamics, and the broader channels through which monetary policy affects the economy—making it an analytical framework well suited for the Prudent Risk Management Approach.

QPM Model

The Quarterly Projection Model (QPM) has long been the traditional workhorse of FPAS Mark I institutions operating under inflation targeting. While ENDOCRED now serves as the primary policy model at the CBA, the QPM continues to play an important complementary role within the analytical toolkit. As a linear semi-structural model, QPM is particularly useful for filtering historical data and for generating internally consistent estimates of key unobservable variables. These include the neutral interest rate, the real effective exchange rate gap, and the output gap—core indicators needed to assess the cyclical position of the economy and to anchor policy discussions.

GPM Model

The Global Projection Model (GPM) also remains a valuable part of the CBA's modeling architecture, particularly for analyzing Armenia's external environment. GPM is a three-country model that captures the economies most relevant for Armenia: the United States, the euro area, and Russia. The model generates coherent projections for these major partners, allowing staff to assess external demand, global inflationary pressures, and interest-rate dynamics that shape Armenia's open-economy conditions.

In addition to its country blocks, GPM incorporates commodity prices that are especially important for Armenia's macroeconomic outlook. These include copper—one of Armenia's significant export products—as well as food and oil, both of which play sizeable roles in domestic inflation, given Armenia's reliance on imported food and its

status as an oil importer. By providing structured projections for these external drivers, GPM offers a clear frame of reference for understanding external risks and evaluating how global developments propagate into the domestic economy under the FPAS Mark II scenario-based framework.

5. Analytical Dashboards

High-quality current economic analysis is the foundation of the Monetary Policy Department's work. It is conducted continuously and presented to the Board during weekly analytical briefings every Wednesday, which cover all policy-relevant developments in both the global and domestic economy. To strengthen efficiency, consistency, and analytical rigor, the Monetary Policy department has developed a series of technical tools—foremost among them, the analytical dashboards.

These dashboards cover a wide range of economic themes: the labor market, inflation dynamics, external developments, sectoral growth patterns, and the decomposition of macroeconomic projections into production and expenditure components, among others. Each dashboard serves as a centralized platform that automatically draws data from the department's unified data warehouse, transforms the raw series into analytically meaningful variables, and presents them in a format designed for fast and rigorous assessment.

The dashboards generate several important benefits. First, they dramatically **increase efficiency**. By automating data extraction and calculation, they eliminate much of the manual work associated with gathering frequently used indicators. This allows staff to devote more time to deeper analysis—whether that involves understanding the drivers of inflation, assessing the main contributors to growth, or conducting ad-hoc investigations during periods of heightened uncertainty.

Second, dashboards greatly facilitate **knowledge transfer** between rotating teams. One of the persistent challenges of a quarterly policy cycle is ensuring continuity when staff teams change. Dashboards provide a unified analytical structure built from common data and standard transformations. New teams can continue the work of their predecessors without having to rebuild datasets or replicate previous calculations, ensuring continuity in both analysis and narrative development.

Finally, dashboards **simplify several complex, recurring tasks** that previously required substantial time and effort. A clear example is the decomposition of GDP growth into expenditure components—an essential step in translating output from the ENDOCRE model into detailed macroeconomic stories. What was once a labor-intensive process is now automated through a dedicated dashboard, saving considerable staff time and

reducing the risk of inconsistencies. Similar efficiencies have been achieved across other standardized processes.

Together, these analytical dashboards create a more efficient, consistent, and resilient analytical environment.

6. Underlying Inflation Measures (NTSPI)

While headline CPI remains the primary benchmark for monetary policy, its individual components provide critical insight for risk-management-based policymaking. Effective policy discussions—and credible communication—require understanding which components of inflation matter most for transmission, for expectations, and for identifying emerging risks. This raises an important analytical question: *how should the CPI be decomposed to help policymakers make better decisions?*

This question was central to the research on **Non-Traded Sticky Price Inflation (NTSPI)**, which designs a CPI decomposition grounded in modern monetary policy transmission. The aim is to separate components driven mainly by supply and exchange-rate movements from those driven by domestic demand and inflation expectations—the latter being essential for evaluating medium-term price stability.

Traditional underlying inflation measures, such as core, median, or trimmed-mean inflation, rely mostly on statistical filters. Their main purpose is to remove volatility, not necessarily to distinguish items based on economic transmission channels. In contrast, New Open Economy Macroeconomics provides a conceptual rationale for classifying CPI components into **flexible** and **sticky** categories.

Flexible prices are those that respond quickly to shocks—especially exchange-rate movements and supply-side disturbances. These are typically internationally traded goods. Sticky prices, by contrast, adjust more slowly. They often reflect domestic services and non-traded goods, where price-setting is less frequent and more influenced by demand conditions and expectations.

This distinction traces back to the classic Dornbusch overshooting model, where asset prices move quickly while most goods prices adjust sluggishly. Traded goods fall into the flexible bucket because they respond almost immediately to exchange-rate changes. Non-traded goods—such as housing, health, education, and many services—fall into the sticky bucket precisely because their prices adjust more gradually.

Understanding the behavior of these two buckets is essential. Sticky prices respond more slowly to economic conditions, making them informative about the persistence of inflation and the anchoring of expectations. Flexible prices, on the other hand, move

quickly in response to idiosyncratic or external shocks. While these shocks may be temporary, they can signal emerging inflationary pressures—particularly when many flexible components start rising simultaneously due to generalized excess demand.

If monetary policy fails to respond in time, these pressures may spill over into sticky components such as wages and services. Once sticky prices begin to accelerate, reversing the trend becomes much more costly, often requiring a significantly more aggressive tightening that imposes unnecessary harm on the real economy.

The underlying challenge is that CPI aggregates combine prices from diverse markets. Inflation becomes meaningful for policy when many markets collectively indicate excess demand—not when one item is volatile. A decomposition grounded in economic transmission helps policymakers distinguish between transient volatility and genuine underlying pressure.

In Armenia, the NTSPI basket includes non-traded goods and services such as housing, healthcare, education, utilities, transportation, and communication services. As of 2023, these items make up roughly **15 percent** of the Armenian CPI. Monitoring this basket provides valuable information about domestic inflation momentum, expectations, and the degree to which external shocks may be feeding into more persistent components.

7. Integrating Judgment and Model-Based Evidence

FPAS Mark II emphasizes structured judgment - expressed through scenario design, risk mapping, credibility assessments, and interpretation of model outputs. Judgment does not override models but complements them. Staff must explain how judgment modifies model projections, increasing transparency and strengthening policy briefings.

8. Implications for Small Open Economies

Armenia is highly exposed to external shocks such as commodity prices, global markets, and geopolitical developments. FPAS Mark II equips policymakers to analyse nonlinearities, asymmetries, and risk distributions more effectively critical tools for small open economies.

By reducing vulnerability to forecast errors and improving resilience to external shocks, FPAS Mark II strengthens macroeconomic stability.

Conclusion

The Analytical Toolkit of FPAS Mark II transforms forecasting and policymaking into a structured, nonlinear, credibility-aware learning system. By integrating models, dashboards, improved measures of underlying inflation, and structured judgment, FPAS

Mark II enables least regrets decision-making under uncertainty and enhances institutional credibility.

Literature & Further Reading

- Adrian T, D. Laxton, and M. Obstfeld 2018, "Advancing the Frontiers of Monetary Policy."
- Alich, A., '2015, "A New Methodology for Estimating the Output Gap in the United States," IMF Working Paper No. 15/144.
- Alich, A., O. Binizima, D. Laxton, K. Tanyeri, H. Wang, J. Yao, and F.Zhang, 'Tᄁᄁ, "Multivariate Filter Estimation of Potential Output for the United States," IMF Working Paper No. 17/106.
- Alich, A., Avetisyan, H., Laxton, MD., Mkhattrishvili, S., Nurbekyan, A., Torosyan, L., Wang, H., "Multivariate Filter Estimation of Potential Output for the United States: An Extension with Labor Market Hysteresis," IMF Working Paper No. 19/35.
- Alich, A., R. Al-Mashat, H. Avetisyan, J. Benes, O. Bizimana, A. Butavyan, R. Ford, N. Ghazaryan, V. Grigoryan, M. Harutyunyan, A. Hovhannisyan, E. Hovhannisyan, H. Karapetyan, M. Kharaisvili, D. Laxton, A. Liqokeli, K. Matikyan, G. Minasyan, S. Mkhattrishvili, A. Nurbekyan, A. Orlov, B. Pashinyan, G. Petrosyan, Y. Rezepina, A. Shirkhanyan, T. Sopromadze, L. Torosyan, E. Vardanyan, H. Wang, and J. Yao, 2018, "Estimates of Potential Output and the Neutral Rate for the U.S. Economy." IMF Working Paper No. 18/152.
- Al-Mashat, R., A. Bulíř, N.N. Dinçer, T. Hlédik, T. Holub, A. Kostanyan, D. Laxton, A. Nurbekyan, and H. Wang, 2018, "An Index for Transparency for Inflation-Targeting Central Banks. Application to the Czech National Bank." IMF Working Paper No. 18/210.
- Laxton, D., Galstyan, M., and Avagyan, V. January 2024. Prudent Risk anagement Approach to Monetary Policy: Theory and Practice of FPAS Mark II at the Central Bank of Armenia.
- Laxton, D., Avetisyan, H. and Avagyan, V.. December 2025. Prudent Risk Management Approach to Price Stability: Theory and Practice of FPAS Mark II at the Central Bank of Armenia. Unpublished Working Draft.

- Freedman, C., & Laxton, D. (2009). “Why Inflation Targeting?” IMF Working Paper 09/86.
- Debelle, G., & Laxton, D. (1996). “Is the Phillips Curve Really a Curve? Some Evidence for Australia.” IMF Working Paper 96/37.
- Kostanyan, A., Matinyan, A., & Papikyan, A. (2022). *Getting FIT with Imperfect Policy Credibility: DYNARE/JULIA Workshops with an Application for the U.S. Economy*. CBA Working Paper 2022/04. October 31, 2022.
- Kostanyan, A., Matinyan, A., & Papikyan, A. (2022). *Getting FIT with Imperfect Policy Credibility: DYNARE/JULIA Workshops with an Application for a Small Open Economy*. CBA Working Paper 2022/07. November 30, 2022.
- Berg, A., P. Karam, and D. Laxton (2006), “A Practical Model-Based Approach to Monetary Policy Analysis—Overview,” IMF Working Paper 06/80.
- Laxton D., A. Kostanyan, A. Liqokeli, G. Minasyan, A. Nurbekyan and T. Sopromadze, 2019, "Mind the Gaps! Financial-Cycle Output Gaps and Monetary-Policy-Relevant Output Gaps." London School of Economics and Political Science.
- CBA Academy. 2025. *Onboarding Platform*. Central Bank of Armenia, December 2025.

Script

Hello everyone, I’m Susanna Grigoryan from the Central Bank of Armenia and a Level One student at the Global Forecasting School. In this session, we’ll look at how the analytical toolkit of FPAS Mark II has evolved, and how it helps us make better policy decisions under uncertainty.

Under FPAS Mark I, most central banks relied on a deterministic baseline and a single linear core model. That framework was extremely valuable for transparency and for building modern inflation-targeting regimes. At the CBA, as in many other central banks, the Quarterly Projection Model introduced by Berg, Karam, and Laxton became the main workhorse. It combined New Keynesian features like rigidities and rational expectations with a policy rule that adjusted interest rates to stabilize inflation and anchor expectations. This was a major step forward when inflation targeting was introduced in Armenia in 2006.

But the world has changed. Since the onset of the COVID-19 pandemic, the global policy environment has been shaped by an very high and persistent level of uncertainty. Alan Greenspan once called uncertainty the “defining feature of the monetary policy landscape,” and recent years have made that description even more accurate. We have seen overlapping shocks from pandemics, geopolitical tensions, energy price swings, and supply-chain disruptions. The global economy has become more susceptible to nonlinear dynamics and high-impact tail events, and this has exposed the limits of relying on a single linear baseline projection.

Traditional FPAS frameworks assume linear relationships and treat credibility as something fixed in the background. In reality, credibility is earned and lost over time. When the output gap is strongly positive, for example, the Phillips curve can steepen, and inflation expectations can accelerate faster than linear models would suggest. If policy reacts too late, the adjustment costs for inflation and output become much higher. The same logic applies to credibility: once it starts to erode, rebuilding it is slow and costly. In this kind of world, a framework that ignores nonlinearities and credibility dynamics underestimates the true risks of policy mistakes.

These developments were a major catalyst for FPAS Mark II. The new framework recognizes that uncertainty is structural and that risk management—not point prediction—must guide policy. That means we need tools that can help build realistic scenarios, test sensitivities, and assess how different policy paths affect credibility over time. Nonlinear structures and explicit risk management are no longer optional add-ons; they are central to how we think about policy.

FPAS Mark II therefore incorporates nonlinearities and state dependence directly into the analytical core. Inflation does not respond in the same way in every state of the world. In overheating conditions, when the output gap is large and positive and expectations are at risk of de-anchoring, inflation responds much more strongly to demand than it does in periods of slack. A nonlinear Phillips curve captures this reality and shows why late tightening can be far more costly than early and measured action. The same applies to financial conditions, transmission channels, and credibility: the impact of a small policy error is not constant across states of the world.

Credibility itself is treated as an endogenous variable. In FPAS Mark II, credibility is modeled as a stock that evolves over time with the central bank’s actions. It depends on the ability to achieve the inflation target, on the consistency of decisions, on the quality of communication, and on institutional behavior more broadly. As the central bank demonstrates competence and reliability, expectations adjust and credibility strengthens. When policy is inconsistent or poorly explained, credibility weakens.

Treating credibility this way tightens the link between today’s decisions and tomorrow’s outcomes and reinforces the idea that credibility must be earned continuously, not assumed.

To support this way of thinking, FPAS Mark II does not rely on a single model. It uses a suite of complementary models, each serving a specific purpose. The ENDOCRED model is at the center. It builds on the traditional QPM but endogenizes credibility, introduces a convex Phillips curve, and extends the IS curve to include a long-term effective interest rate derived from the yield curve and a time-varying term premium. It uses a loss-function-based policy design instead of a simple Taylor rule, and it explicitly links dollarization to credibility, so that as credibility improves, dollarization declines and the effectiveness of domestic monetary policy increases.

Alongside ENDOCRED, the CBA uses other models that play distinct roles. MPNOD helps estimate latent variables such as the output gap and neutral interest rate by linking unobservable concepts to observed data. A financial-cycle model-FCMOD-focuses on credit, leverage, and systemic vulnerabilities. The linear QPM is retained as a complementary tool for filtering history and extracting consistent estimates of unobservables, especially useful for historical decompositions and pedagogical comparison. A monthly projection model helps think about near-term scenarios using higher-frequency data. DSGE-type models, such as GIMF-style structures, are used for medium-term policy questions where fiscal and monetary interactions are central. And the Global Projection Model provides a structured representation of Armenia’s main external partners—the United States, the euro area, and Russia—as well as key commodities like oil, copper, and food, all of which are crucial for a small open economy like Armenia.

Using a model suite spreads analytical risk. No model is treated as the “truth.” Instead, each model provides a different lens, and enhances policy discussions.

The analytical infrastructure supporting these models is just as important. High-quality current analysis is carried out continuously and presented to the Board every Wednesday. To make this work more efficient and consistent, the Monetary Policy Department relies heavily on analytical dashboards. These dashboards draw automatically from a unified data warehouse and transform raw series into meaningful indicators across key areas—labor markets, inflation, external conditions, sectoral growth, and the breakdown of GDP into expenditure and production components.

By automating data collection and transformation, dashboards drastically reduce the time staff spend on repetitive work and free them to focus on interpretation and risk

assessment. They also make team rotations much smoother. When staff move between teams, the dashboards ensure that data, methods, and key indicators are consistent over time, so new teams can pick up where previous teams left off without rebuilding everything from scratch. Complex recurring tasks, such as decomposing GDP growth coming from ENDOCRED into detailed demand-side stories, are now managed through dedicated dashboards instead of manual spreadsheets. Beyond efficiency, dashboards integrate high-frequency indicators, model outputs, financial conditions, sectoral data, risk distributions, and scenario trees. They help staff present risk-aware assessments and allow the Board to visualize uncertainty in a structured and intuitive way.

Another important improvement in the FPAS Mark II toolkit is the treatment of underlying inflation. Headline CPI is still the main policy benchmark, but its components carry crucial information for risk management. In Armenia, the Non-Traded Sticky Price Inflation index—NTSPI—has been developed to distinguish between flexible and sticky components in a way that reflects the transmission mechanism. Flexible prices, typically found in traded goods, adjust quickly to exchange-rate and supply shocks. Sticky prices, often in non-traded services like housing, healthcare, and education, adjust slowly and are more closely tied to domestic demand and expectations. Monitoring NTSPI, which covers non-traded goods and services such as utilities, transportation, communication, and other services that make up around 15 percent of the CPI basket, gives policymakers a clearer sense of persistent inflation pressures and whether temporary shocks are spilling over into components that are harder to reverse.

All of this feeds into a broader principle: FPAS Mark II is not about replacing judgment with models; it is about structuring judgment. Judgment appears in the design of scenarios, in the mapping of risks, in the interpretation of model outputs, and in the assessment of credibility. Staff are expected to explain how and why their judgment modifies model projections, making the reasoning behind recommendations transparent. The Board, in turn, uses this structured analysis to apply least-regrets thinking when it votes on policy.

For a small open economy like Armenia, exposed to external shocks, commodity price swings, and geopolitical developments, this approach is especially important. FPAS Mark II equips policymakers to analyze nonlinearities, asymmetries, and risk distributions more effectively. It reduces vulnerability to forecast errors and strengthens the institution's ability to navigate external shocks without over-reacting or under-reacting.

In summary, the analytical toolkit of FPAS Mark II turns forecasting and policymaking into a structured, nonlinear, credibility-aware learning system. By combining model suites, dashboards, improved measures of underlying inflation, and structured judgment, FPAS

Mark II supports least-regrets decision-making under uncertainty and helps sustain institutional credibility over time.

Thank you for watching, and I would like to acknowledge Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervolidze at the Global Forecasting School for their guidance and support in producing this video.

See you in the next video!